RANDOMIZED ROBUST PRICE OPTIMIZATION

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Pricing problem

- I: number of products, indexed by $i = 1, \ldots, I$
- p_i: price of product i
- $\mathbf{p} = (p_1, p_2, \ldots, p_I)$: vector of prices
- $\mathcal{P}$: set of feasible price vectors $\mathbf{p}$
- $d_i(\mathbf{p})$: demand of product i as a function of price vector $\mathbf{p}$
- $R(\mathbf{p})$: revenue function, defined as

$$R(\mathbf{p}) = \sum_{i=1}^{I} p_i \cdot d_i(\mathbf{p})$$

- (Nominal) multi-product pricing problem:

$$\max_{\mathbf{p} \in \mathcal{P}} R(\mathbf{p})$$

Demand models

Examples of demand models:

- Linear demand:

$$d_i(\mathbf{p}) = \alpha_i - \beta_i p_i + \sum_{j \neq i} \gamma_{i,j} p_j$$

- Semi-log demand: log of demand is linear in prices:

$$\log d_i(\mathbf{p}) = \alpha_i - \beta_i p_i + \sum_{j \neq i} \gamma_{i,j} p_j$$

  which means

$$d_i(\mathbf{p}) = e^{\alpha_i - \beta_i p_i + \sum_{j \neq i} \gamma_{i,j} p_j}$$

- Log-log demand: log of demand is linear in logs of prices:

$$\log d_i(\mathbf{p}) = \alpha_i - \beta_i \log p_i + \sum_{j \neq i} \gamma_{i,j} \log p_j$$

  which means

$$d_i(\mathbf{p}) = e^{\alpha_i} p_i^{-\beta_i} \cdot \prod_{j \neq i} p_j^{\gamma_{i,j}}$$

Model uncertainty

- Typically, we assume that the demand functions $d_1(\cdot), \cdot, d_I(\cdot)$ are known
- In practice, we do not know these demand functions exactly and must estimate them from limited data
- Challenge: how do we account for uncertainty in demand functions?
  - If we assume $d_1, \ldots, d_I$ and find price vector $\mathbf{p}^*$, but actual demand is given by $\tilde{d}_1, \ldots, \tilde{d}_I$, then $\mathbf{p}^*$ could actually be highly suboptimal and perform poorly

Deterministic robust optimization

- In OR, a leading paradigm for managing uncertainty is robust optimization
- Instead of solving the problem $\max_{x \in \mathcal{X}} f(x, u)$ for a fixed estimate of u, we solve

$$\max_{x \in \mathcal{X}} \min_{u \in \mathcal{U}} f(x, u)$$

  where $\mathcal{U}$ is an uncertainty set
- Huge literature:
  - Ben Tal et al. (2009), Bertsimas et al. (2011), Gabrel et al. (2014), Bertsimas and den Hertog (2022)
  - Applications in inventory management (Ardestani-Jaafari and Delage 2016, Bertsimas and Thiele 2006, Govindarajan et al. 2021), assortment optimization (Rusmevichientong and Topaloglu 2012, Bertsimas and Mišić 2017, Sturt 2024), finance (Delage and Ye 2010, Bandi and Bertsimas 2014, Sturt 2023), ...

Deterministic robust optimization

- Deterministic robust price optimization (DRPO) problem:

$$\text{DRPO}: \max_{\mathbf{p}\in\mathcal{P}} \min_{\mathbf{u}\in\mathcal{U}} R(\mathbf{p}, \mathbf{u}),$$

  where $\mathcal{U}$ is uncertainty set of demand model parameters and R is function of both price vector $\mathbf{p}$ and demand model parameters $\mathbf{u}$

Randomized robust optimization

Recently, there has been increasing interest within the robust optimization community on *randomized* decisions:

- We choose a distribution over a set of actions
- Adversary/nature only knows the distribution but not the action that we take
- By randomizing, it is possible to achieve better worst-case objectives than a fixed deterministic decision

A growing literature:

- Delage, Kuhn and Wiesemann (2018), *MS*: "Dice"-sion–making under uncertainty: When can a random decision reduce risk?
- Other work: Mastin et al. (2015), Bertsimas et al. (2016), Delage and Saif (2022), Sadana and Delage (2023), Wang et al. (2024)

Randomized robust optimization

- Randomized robust price optimization (RRPO) problem:

$$\text{RRPO}: \quad \max_{F \in \mathcal{F}} \min_{\mathbf{u} \in \mathcal{U}} \int_{\mathcal{P}} R(\mathbf{p}, \mathbf{u})\, dF(\mathbf{p}),$$

- When $\mathcal{P}$ is discrete, we instead write:

$$\text{RRPO-D}: \quad \max_{\boldsymbol{\pi} \in \Delta_{\mathcal{P}}} \min_{\mathbf{u} \in \mathcal{U}} \sum_{\mathbf{p} \in \mathcal{P}} \pi_{\mathbf{p}} R(\mathbf{p}, \mathbf{u}).$$

where Δ_S is the $(|S| - 1)$-dimensional unit simplex

Implementing random prices

- Suppose we have three price vectors: $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$
- Suppose $\pi_{\mathbf{p}_1} = 0.2, \pi_{\mathbf{p}_2} = 0.3, \pi_{\mathbf{p}_3} = 0.5$

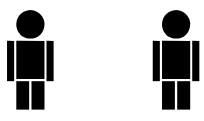
Randomize by arriving customer (ecommerce):



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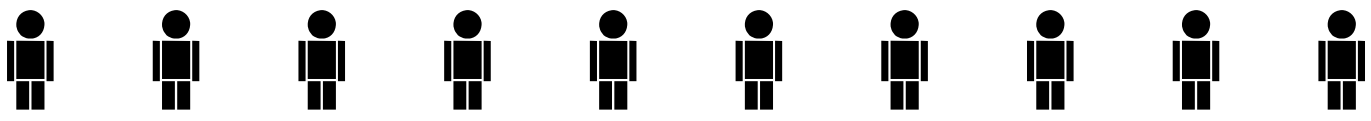
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Contributions

1. **Benefit of randomization**: We analyze when randomized robust pricing provides a benefit over deterministic robust pricing, and develop conditions when the two approaches are equivalent in worst-case revenue.
2. **Methodology**: We develop exact solution methods for solving the randomized robust pricing problem when $\mathcal{P}$ is finite, and $\mathcal{U}$ is either convex or finite.
3. **Numerical experiments**: We compare the randomized and deterministic robust approaches using synthetic and real data, and show that randomized pricing can achieve significantly higher worst-case and out-of-sample revenues than deterministic robust pricing.

When is randomization beneficial?

Randomization-proof versus randomization-receptive

- Z_{DR}: optimal objective of deterministic robust price optimization problem
- Z_{RR}: optimal objective of randomized robust price optimization problem
- We say a problem is *randomization-proof* if $Z^*_{\mathrm{RR}} = Z^*_{\mathrm{DR}}$
- Otherwise, we say a problem is *randomization-receptive* if $Z^*_{\mathrm{RR}} > Z^*_{\mathrm{DR}}$

Example: quasiconcavity/quasiconvexity of revenue function

Theorem

Suppose that $\mathcal{P} \subseteq \mathbb{R}^I$ *and* $\mathcal{U} \subseteq \mathbb{R}^M$ *are compact convex sets. Suppose that* $R(\mathbf{p}, \mathbf{u})$ *is quasiconcave in* $\mathbf{p}$ *on* $\mathcal{P}$ *for any* $\mathbf{u} \in \mathcal{U}$*, quasiconvex in* $\mathbf{u}$ *on* $\mathcal{U}$ *for any* $\mathbf{p} \in \mathcal{P}$ *and continuous in* $(\mathbf{p}, \mathbf{u})$*. Then* $Z^*_{\mathrm{DR}} = Z^*_{\mathrm{RR}}$*.*

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Proof sketch:

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Proof sketch:

$$Z^*_{\mathrm{RR}} = \max_{F \in \mathcal{F}} \min_{\mathbf{u} \in \mathcal{U}} \int_{\mathcal{P}} R(\mathbf{p}, \mathbf{u}) \, dF(\mathbf{p})$$

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Z^*_{\mathrm{RR}} &= \max_{F \in \mathcal{F}} \min_{\mathbf{u} \in \mathcal{U}} \int_{\mathcal{P}} R(\mathbf{p}, \mathbf{u}) \, dF(\mathbf{p}) \\
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Consequences of results

- Other omitted results: randomization-proofness occurs if:
  - Revenue function is concave for every $\mathbf{u}$ and $\mathcal{P}$ is convex
  - Strong duality holds for finite $\mathcal{P}$
- These results allow us to show that some robust pricing problems are randomization-proof:
  - E.g., semi-log demand or log-log demand model with one product is in general randomization-proof
  - Linear demand model with multiple products is randomization-proof if elasticity matrix is always negative semidefinite
- In general though, conditions are restrictive:
  - In the multiproduct case, semi-log and log-log demand models lead to revenue functions that are not concave or quasiconcave in $\mathbf{p}$
  - In general when $\mathcal{P}$ is discrete, strong duality does not hold

Solution methodology

Solution methodology for finite $\mathcal{P}$, convex $\mathcal{U}$

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- RRPO problem can be re-formulated as follows:

$$\max_{\boldsymbol{\pi} \in \Delta_{\mathcal{P}}} \min_{\mathbf{u} \in \mathcal{U}} \sum_{\mathbf{p} \in \mathcal{P}} \pi_{\mathbf{p}} R(\mathbf{p}, \mathbf{u})$$

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Epigraph form of last problem:

$$\begin{aligned} \underset{t,\mathbf{u}}{\text{minimize}} \quad & t \\ \text{subject to} \quad & t \geq R(\mathbf{p},\mathbf{u}), \quad \forall\ \mathbf{p}\in\mathcal{P}, \\ & \mathbf{u}\in\mathcal{U}. \end{aligned}$$

Solving the dual problem

Dual problem:

$$\begin{aligned} \underset{t,\mathbf{u}}{\text{minimize}} \quad & t \\ \text{subject to} \quad & t \geq R(\mathbf{p}, \mathbf{u}), \quad \forall\ \mathbf{p} \in \mathcal{P}, \\ & \mathbf{u} \in \mathcal{U}. \end{aligned}$$

- When $R(\mathbf{p}, \mathbf{u})$ is convex in $\mathbf{u}$, this is a convex program, but with combinatorially many constraints
  - Solve using constraint generation
- Success of this critically depends on being able to solve the separation problem

$$\max_{\mathbf{p} \in \mathcal{P}} R(\mathbf{p}, \mathbf{u})$$

- We will assume that $\mathcal{P} = \mathcal{P}_1 \times \cdots \times \mathcal{P}_I$, where each $\mathcal{P}_i$ is a finite set

Convex $\mathcal{U}$, finite $\mathcal{P}$: semi-log demand model

- For the semi-log demand model, revenue function is

$$R(\mathbf{p}, \mathbf{u}) = \sum_{i=1}^{I} p_i \cdot e^{\alpha_i - \beta_i p_i + \sum_{j \neq i} \gamma_{i,j} p_j}$$

- R is convex in $\mathbf{u}$
- Separation problem is

$$\max_{\mathbf{p} \in \mathcal{P}} \sum_{i=1}^{I} p_i \cdot e^{\alpha_i - \beta_i p_i + \sum_{j \neq i} \gamma_{i,j} p_j}$$

- R is not concave in $\mathbf{p}$; not log-concave or even quasi-concave...
- ... can we solve it?

Solving the semi-log separation problem

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- The optimal solution set is unchanged if we consider the log-transformed objective:

$$\max_{\mathbf{p}\in\mathcal{P}} \log \left(\sum_{i=1}^{I} p_i \cdot e^{\alpha_i - \beta_i p_i + \sum_{j\neq i} \gamma_{i,j} p_j} \right)$$

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- Re-write the objective slightly:

$$\max_{\mathbf{p}\in\mathcal{P}} \log \left(\sum_{i=1}^{I} e^{\alpha_i + \log p_i - \beta_i p_i + \sum_{j\neq i} \gamma_{i,j} p_j} \right)$$

Solving the semi-log separation problem

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Recall that for any $\mathbf{y} \in \mathbb{R}^K$,

$$\log\left(\sum_{k=1}^{K} e^{y_k}\right) = \max_{\boldsymbol{\mu} \in \Delta_{[K]}} \left\{ \boldsymbol{\mu}^\top \mathbf{y} - \sum_{k=1}^{K} \mu_k \log \mu_k \right\}.$$

where $\Delta_{[K]} = \{\boldsymbol{\mu} \in \mathbb{R}^K \mid \mathbf{1}^\top \boldsymbol{\mu} = 1, \boldsymbol{\mu} \geq \mathbf{0}\}$.

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where $\Delta_{[K]} = \{\boldsymbol{\mu} \in \mathbb{R}^K \mid \mathbf{1}^\top \boldsymbol{\mu} = 1, \boldsymbol{\mu} \geq \mathbf{0}\}$.
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$$\max_{\mathbf{p} \in \mathcal{P}} \log\left(\sum_{i=1}^{I} e^{\alpha_i + \log p_i - \beta_i p_i + \sum_{j \neq i} \gamma_{i,j} p_j}\right)$$

Solving the semi-log separation problem

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$$= \max_{\mathbf{p} \in \mathcal{P}, \boldsymbol{\mu} \in \Delta_{[I]}} \left\{ \sum_{i=1}^{I} \mu_i (\alpha_i + \log p_i - \beta_i p_i + \sum_{j \neq i} \gamma_{i,j} p_j) - \sum_{i=1}^{I} \mu_i \log \mu_i \right\}$$

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Hence, whole problem becomes:

$$\begin{aligned}
\underset{\mathbf{x},\boldsymbol{\mu}}{\text{maximize}} \quad & \sum_{i=1}^{I} \mu_i \left(\alpha_i + \sum_{t \in \mathcal{P}_i} \log t \cdot x_{i,t} - \beta_i \cdot \sum_{t \in \mathcal{P}_i} t \cdot x_{i,t} + \sum_{j \neq i} \gamma_{i,j} \sum_{t \in \mathcal{P}_j} t \cdot x_{j,t} \right) \\
& - \sum_{i=1}^{I} \mu_i \log \mu_i \\
\text{subject to} \quad & \sum_{i=1}^{I} \mu_i = 1, \\
& \sum_{t \in \mathcal{P}_i} x_{i,t} = 1, \quad \forall\, i \in [I], \\
& x_{i,t} \in \{0,1\}, \quad \forall\, i \in [I],\ t \in \mathcal{P}_i, \\
& \mu_i \geq 0, \quad \forall\, i \in [I].
\end{aligned}$$

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Solving the semi-log separation problem

Introduce $w_{i,j,t}$ as the linearization of $\mu_i \cdot x_{j,t}$; then problem becomes

$$\underset{\boldsymbol{\mu},\mathbf{w},\mathbf{x}}{\text{maximize}} \quad \sum_{i=1}^{I} \mu_i \alpha_i + \sum_{i=1}^{I} \sum_{t \in \mathcal{P}_i} \log t \cdot w_{i,i,t} - \sum_{i=1}^{I} \beta_i \cdot \sum_{t \in \mathcal{P}_i} t \cdot w_{i,i,t}$$

$$+ \sum_{i=1}^{I} \sum_{j \neq i} \gamma_{i,j} \cdot \Big(\sum_{t \in \mathcal{P}_j} t \cdot w_{i,j,t} \Big) - \sum_{i=1}^{I} \mu_i \log \mu_i$$

$$\text{subject to} \quad \sum_{t \in \mathcal{P}_j} w_{i,j,t} = \mu_i, \quad \forall\ i \in [I],\ j \in [I],$$

$$\sum_{i=1}^{I} w_{i,j,t} = x_{j,t}, \quad \forall\ j \in [I],\ t \in \mathcal{P}_j,$$

$$\sum_{i=1}^{I} \mu_i = 1,$$

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$$\mu_i \geq 0, \quad \forall\ i \in [I].$$

Semi-log separation problem

This problem is special for several reasons:

- It is a mixed-integer convex program (convex optimization problem with some variables integer/binary)
- Specifically, it is a mixed-integer exponential cone program
  - As of 2019, Mosek is capable of solving such problems to global optimality via branch-and-bound
- To our knowledge, this is the first formulation of the semi-log multi-product pricing problem as a mathematical program
  - To date, all research that has considered solving this type of problem (in marketing, operations, etc.) has focused on heuristics
- Same technique with application of Sion's minimax theorem can be used to obtain an exact formulation of the DRPO problem
- Same technique also applies to log-log separation problem

Numerical Experiments

Real data experiment

- `orangeJuice`: data set on sales of 11 frozen orange juice products over 83 stores of Dominick's Finer Foods
  - Data accessed via `bayesm` package (Rossi 2012); used previously in Montgomery (1997), Mišić (2020)
- Data consists of weekly prices of products and their sales quantities at different stores
- Set $\mathcal{P}_i$ of each product $i = 1, \ldots, 11$ using 0th, 25th, 50th, 75th, 100th percentiles





Results - semi-log demand model

Uncertainty set:

$$\mathcal{U} = \left\{ \mathbf{u} = (\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}) \,\middle|\, \|\tilde{\mathbf{u}}\|_1 \leq \theta;\ \tilde{\mathbf{u}}_k = \frac{\mathbf{u}_k - \mathbf{u}_{0k}}{\mathbf{u}_{0k}},\ \forall k \in \{1, .., I + I^2\} \right\},$$

θ	t_{RR}	Z^*_{RR}	t_{DR}	Z^*_{DR}	RI(%)
0.10	16.86	342357.06	15.98	290474.67	17.86
0.50	41.42	197517.06	47.65	147748.35	33.68
0.80	42.61	149709.04	67.83	105734.14	41.59
1.00	67.47	125987.02	74.03	86977.24	44.85
1.50	61.70	82880.96	44.65	56474.64	46.76
2.00	65.21	54665.15	52.10	37164.75	47.09

Out-of-sample prescriptions

- Although we have primarily focused on worst-case performance, it is natural to ask whether randomization can lead to prescriptions that are better than the nominal or deterministic robust approaches when compared out-of-sample
- Evaluating out-of-sample prescriptions is (relatively) straightforward for some problems:
  - Newsvendor/inventory management
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Out-of-sample prescriptions for pricing

- For pricing, this is tricky to do:
  - At a given week and a given store, we do not know what sales (and hence revenue) would have been for a different price vector
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  - Solution: consider a pricing problem restricted to historical price vectors - a restricted pricing problem

Data-driven evaluation procedure

- Fix a family of demand models (e.g., log-linear)
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- For that store, set $\mathcal{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_n\}$ to be the unique price vectors that appear over the weeks of data for that store
- Split the set of price vectors as $\mathcal{P} = \mathcal{P}_{\text{train}} \cup \mathcal{P}_{\text{val}} \cup \mathcal{P}_{\text{test}}$

Data-driven evaluation procedure - training

- Use training set price vectors $\mathcal{P}_{\text{train}}$ and their demand data to estimate nominal demand model parameter vector, $\mathbf{u}^{\text{N}}$

Data-driven evaluation procedure - validation

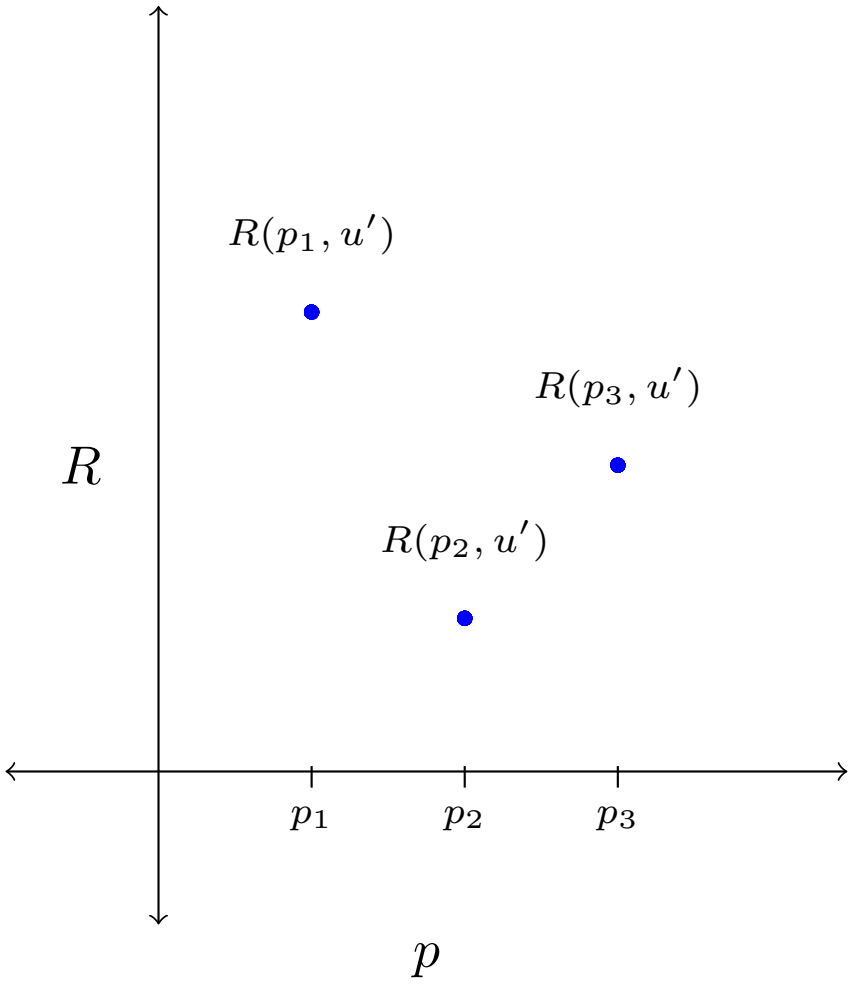
- Use validation set price vectors $\mathcal{P}_{\text{val}}$ and their demand data to tune uncertainty set centered at $\mathbf{u}^N$ for the robust approaches
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  - Evaluate the pricing strategy using the realized demand data
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Data-driven evaluation procedure - testing

- Use test set price vectors $\mathcal{P}_{\text{test}}$ and their demand data to perform final evaluation of DR, RR and N methods:
  - Solve for optimal DR / RR / N pricing strategy restricted to price vectors in $\mathcal{P}_{\text{test}}$
  - Evaluate them using realized demand data for test set

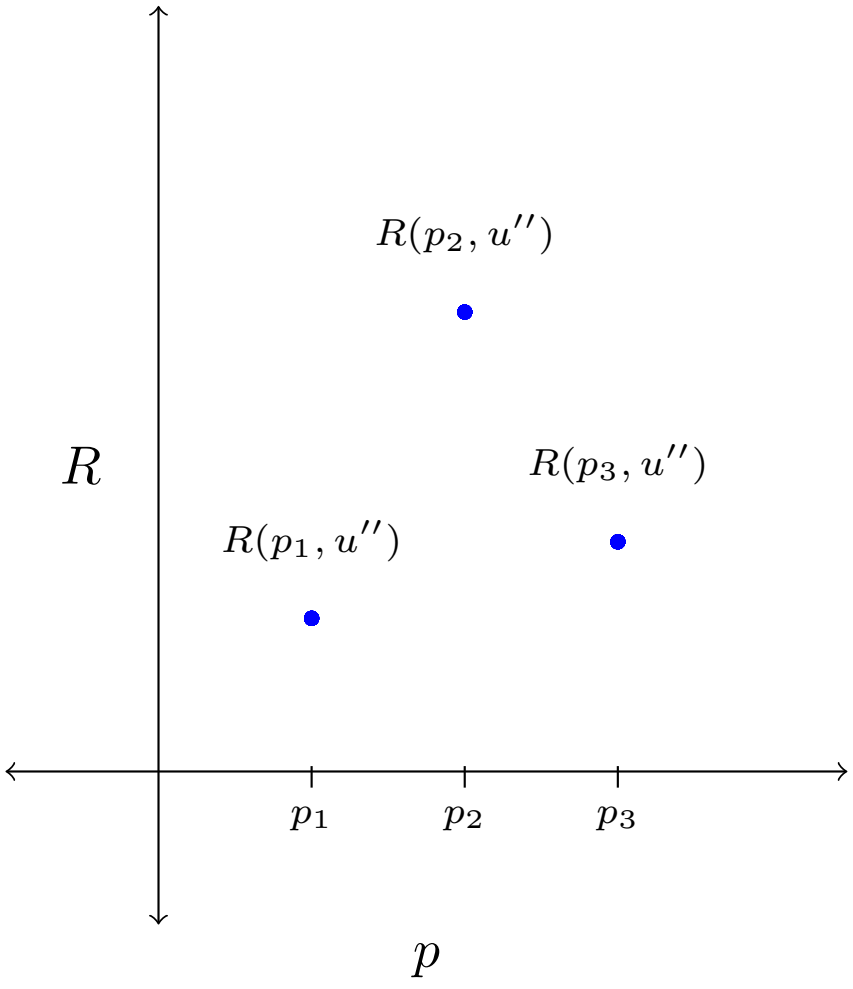
Restricted pricing problem - example

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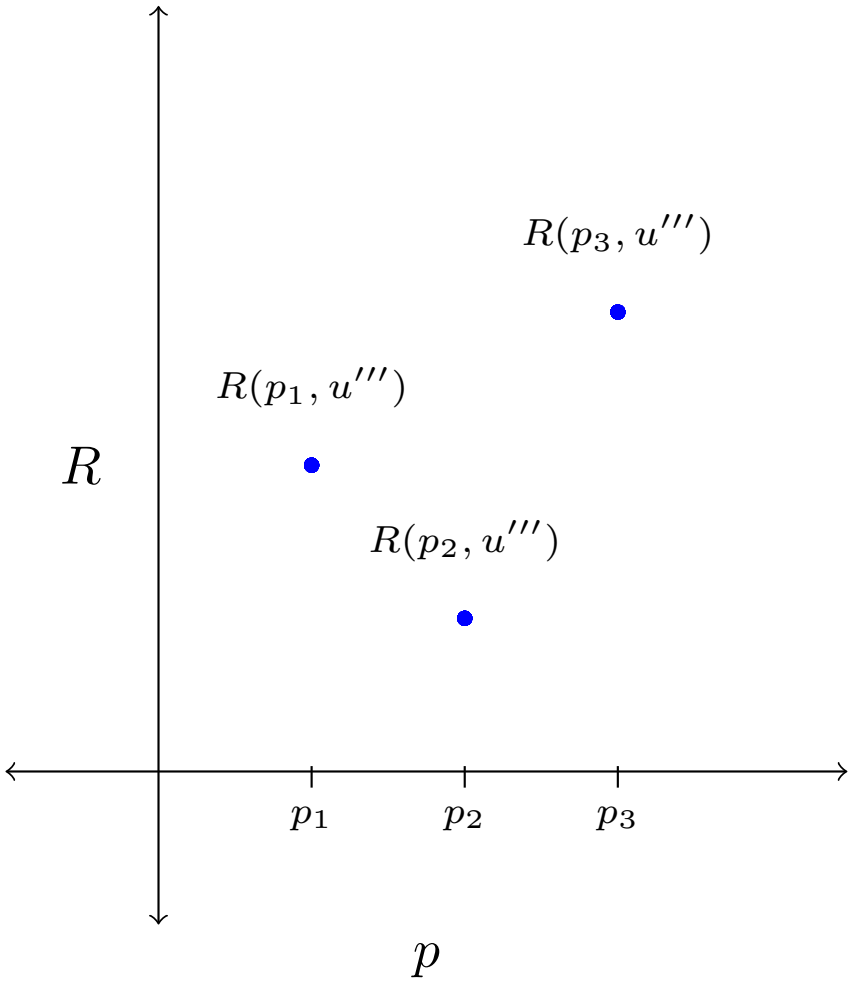
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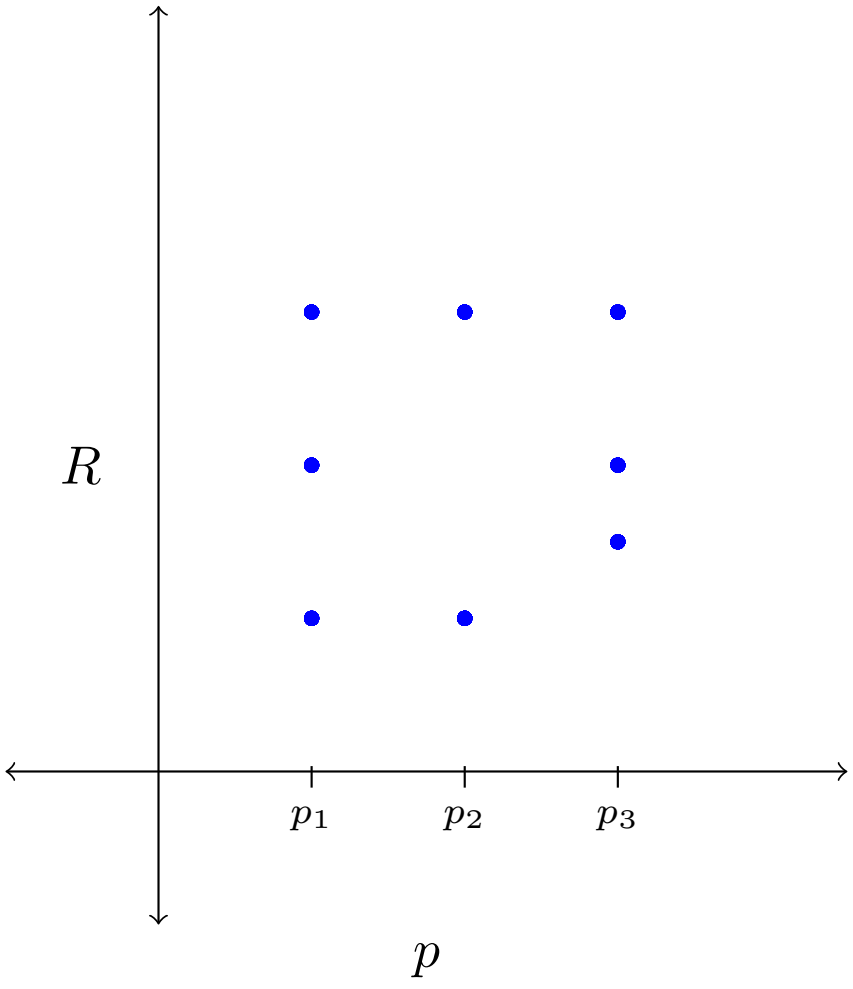
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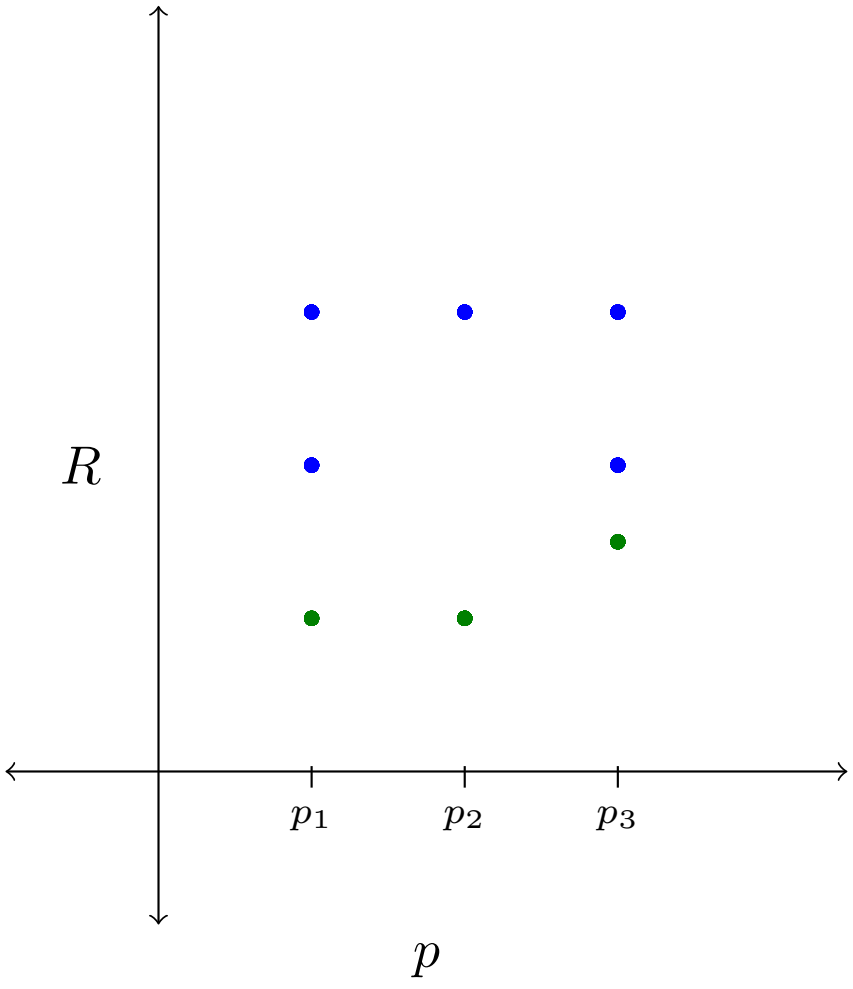
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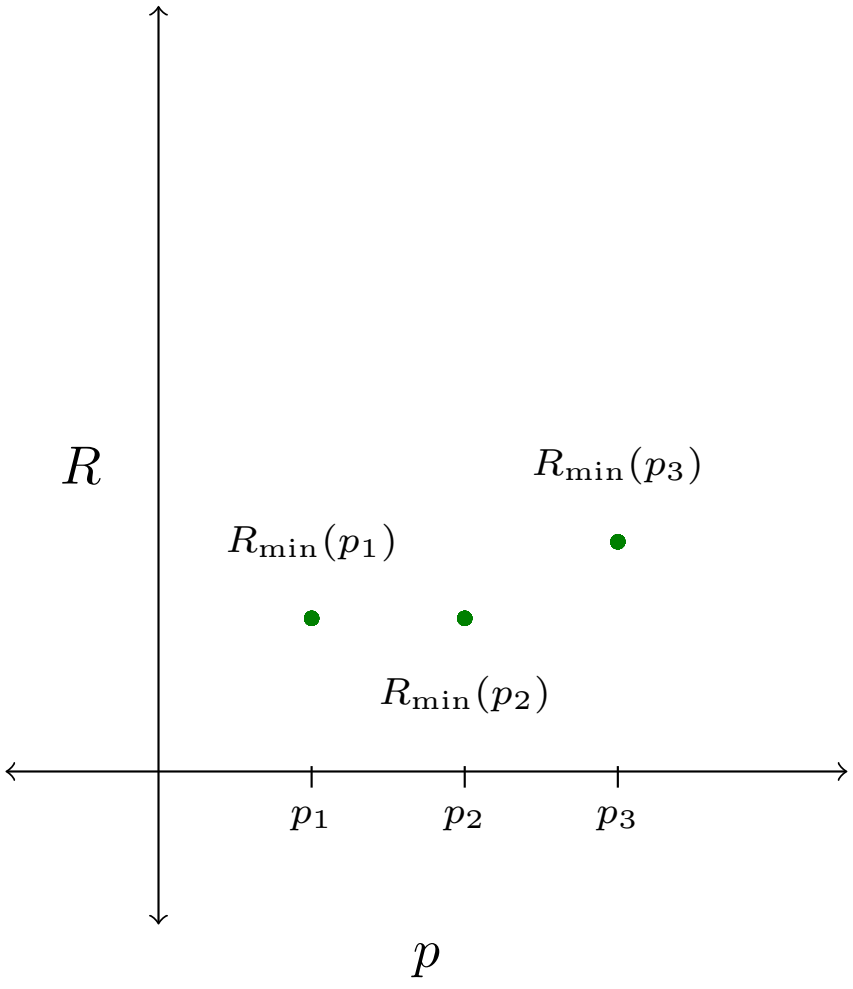
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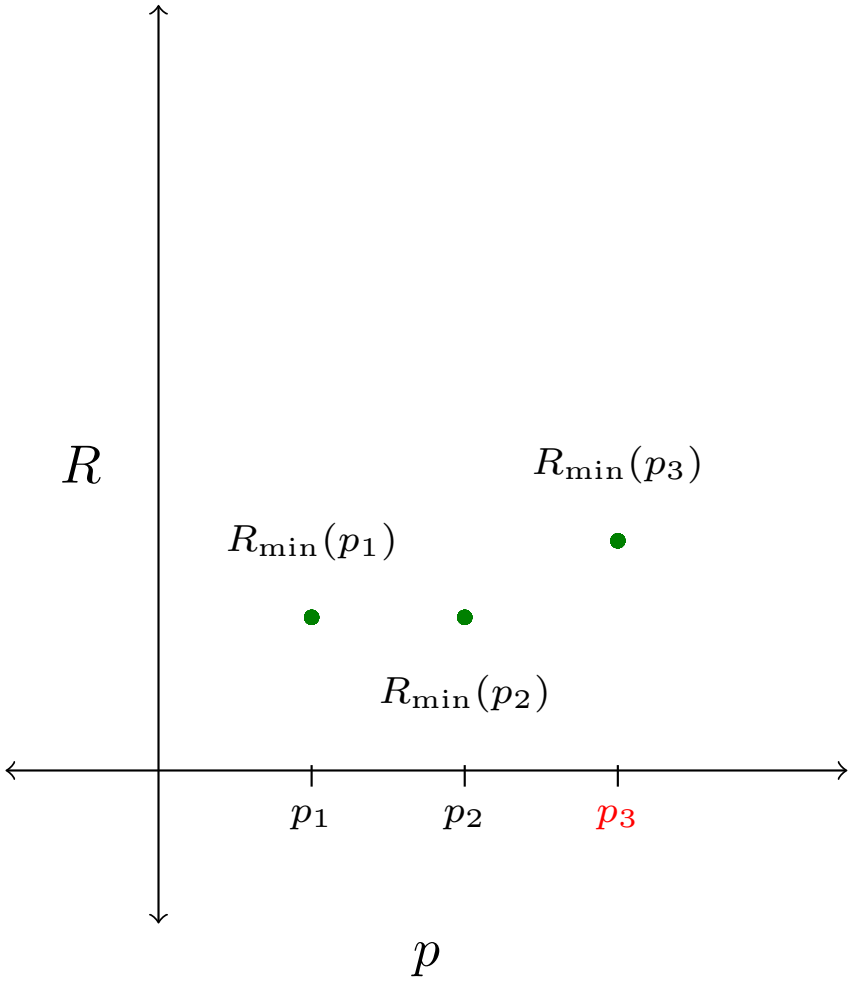
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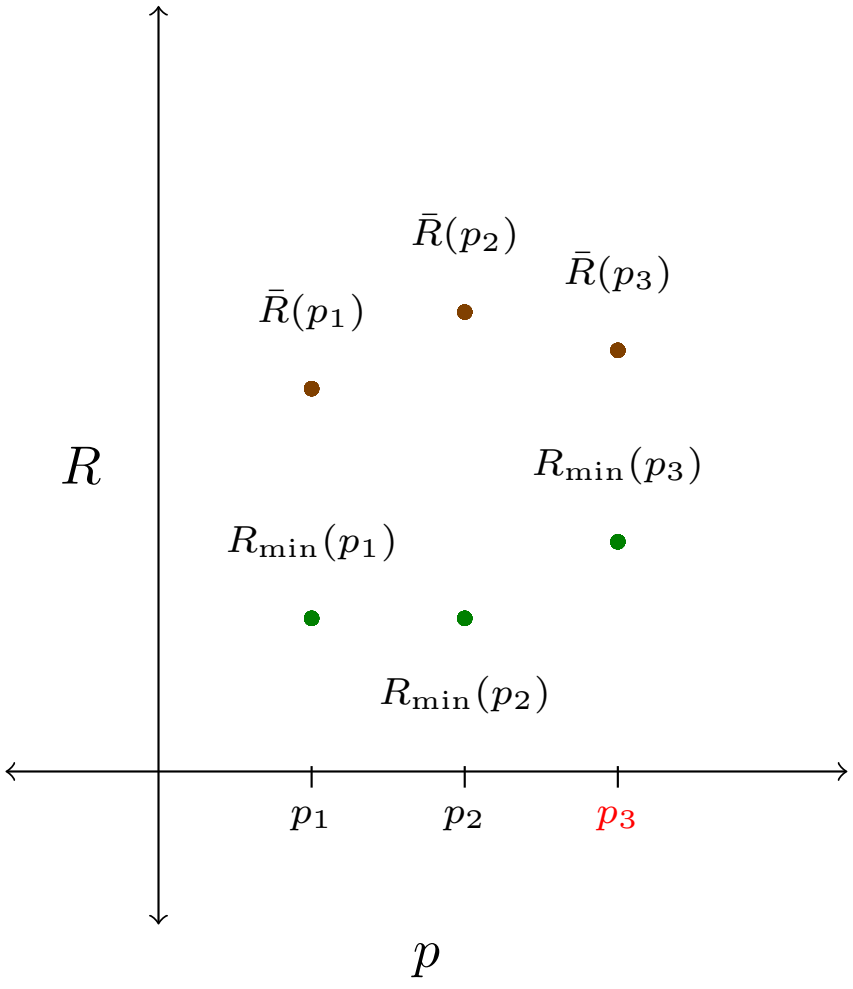
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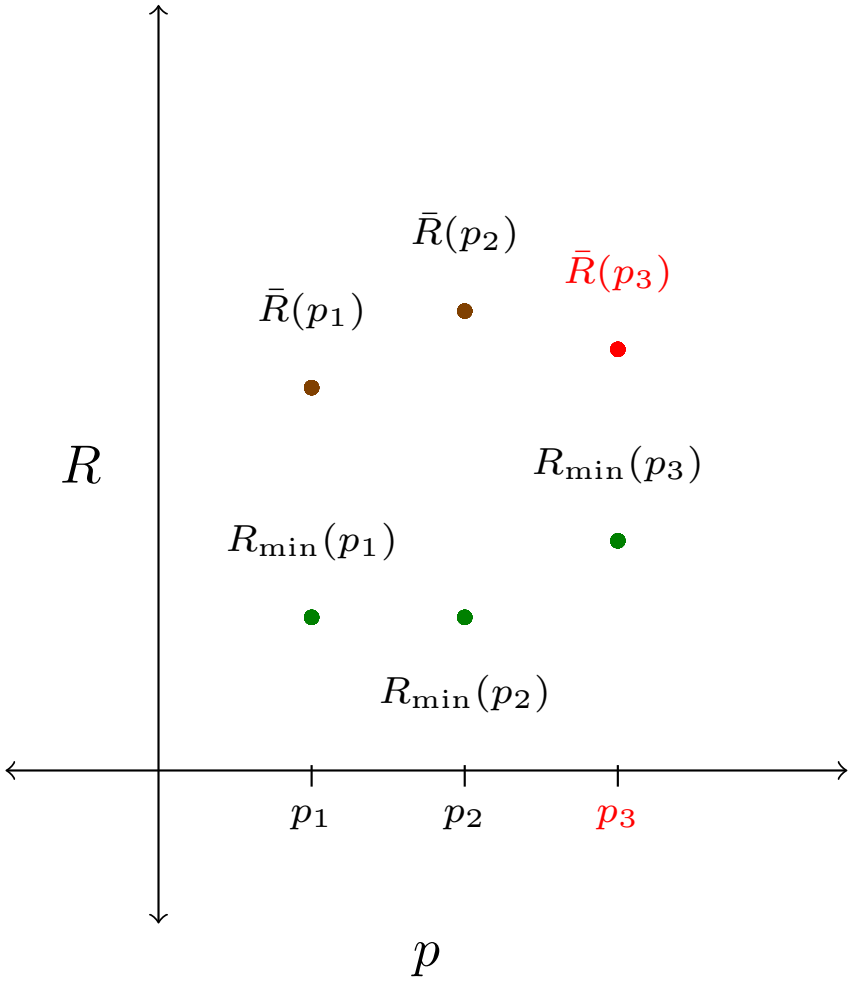
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Results

Demand model	Type of mean	$\mathrm{RI}_{\mathrm{N}\to\mathrm{RR}}$ (%)	$\mathrm{RI}_{\mathrm{N}\to\mathrm{DR}}$ (%)	$\mathrm{RI}_{\mathrm{N}\to\mathrm{H}}$ (%)
Semi-log	Mean	4.65	1.51	3.09
Semi-log	Trimmed Mean (0.05)	1.76	0.17	1.20
Semi-log	Trimmed Mean (0.10)	1.40	0.21	1.08
Log-log	Mean	11.88	14.31	16.61
Log-log	Trimmed Mean (0.05)	8.52	8.53	11.07
Log-log	Trimmed Mean (0.10)	5.37	3.80	6.57

Conclusions

- We develop a pricing methodology for managing demand model uncertainty using randomization
- Using synthetic and real data, we show the benefits of randomized robust pricing over deterministic robust pricing
- Many other directions:
  - How to incorporate covariates/contextual information?
  - How to combine with learning (A/B testing)?
  - How to address strategic behavior?
  - Other revenue management applications where randomization is natural and uncertainty is present (e.g., promotion planning)

Paper

- Guan, X. and Mišić, V. V. (2025) Randomized robust price optimization. Forthcoming in *Management Science*. Available at: `https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4480799`
- Omitted content:
  - Solution methodology for finite $\mathcal{P}$, finite $\mathcal{U}$ (based on Delage and Saif 2022)
  - How to obtain constrained distributions in convex $\mathcal{U}$, finite $\mathcal{P}$ case
  - Conditions under which one can restrict to finite $\mathcal{P}$ (versus convex $\mathcal{U}$)
  - Characterizing the size of the support of $\boldsymbol{\pi}^*$
  - Comparing randomized and deterministic robust prices out-of-sample

Thank you for listening! Questions?

Email: velibor.misic@anderson.ucla.edu
Website: http://vvmisic.github.io/



Additional Material

Complexity of randomization

Complexity of randomization

How big is the support of the optimal randomized robust price prescription?

Complexity of randomization

How big is the support of the optimal randomized robust price prescription?

Theorem

Suppose $R(\mathbf{p}, \mathbf{u})$ *is convex in* $\mathbf{u}$ *for any* $\mathbf{p} \in \mathcal{P}$*, that* $\mathcal{U}$ *is a non-empty convex compact subset of* $\mathbb{R}^M$*, and that* $\mathcal{P}$ *is a finite set. Then there exists an optimal solution* $\boldsymbol{\pi}^*$ *to the RRPO problem* $\max_{\boldsymbol{\pi} \in \Delta_{\mathcal{P}}} \min_{\mathbf{u} \in \mathcal{U}} \sum_{\mathbf{p} \in \mathcal{P}} \pi_{\mathbf{p}} R(\mathbf{p}, \mathbf{u})$*, such that* $\boldsymbol{\pi}^*$ *has a support of at most* $M + 1$*, i.e., the number of price vectors* $\mathbf{p}$ *for which* $\pi_{\mathbf{p}} > 0$ *is at most* $M + 1$*.*

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- Proof involves using the concept of support constraints (Calafiore and Campi 2005) in an algorithmic way
  - Start by solving dual; partition all constraints into non-active constraints, support constraints and active non-support constraints; pick an active non-support constraint and throw it away
  - Keep doing this, until you are only left with support constraints

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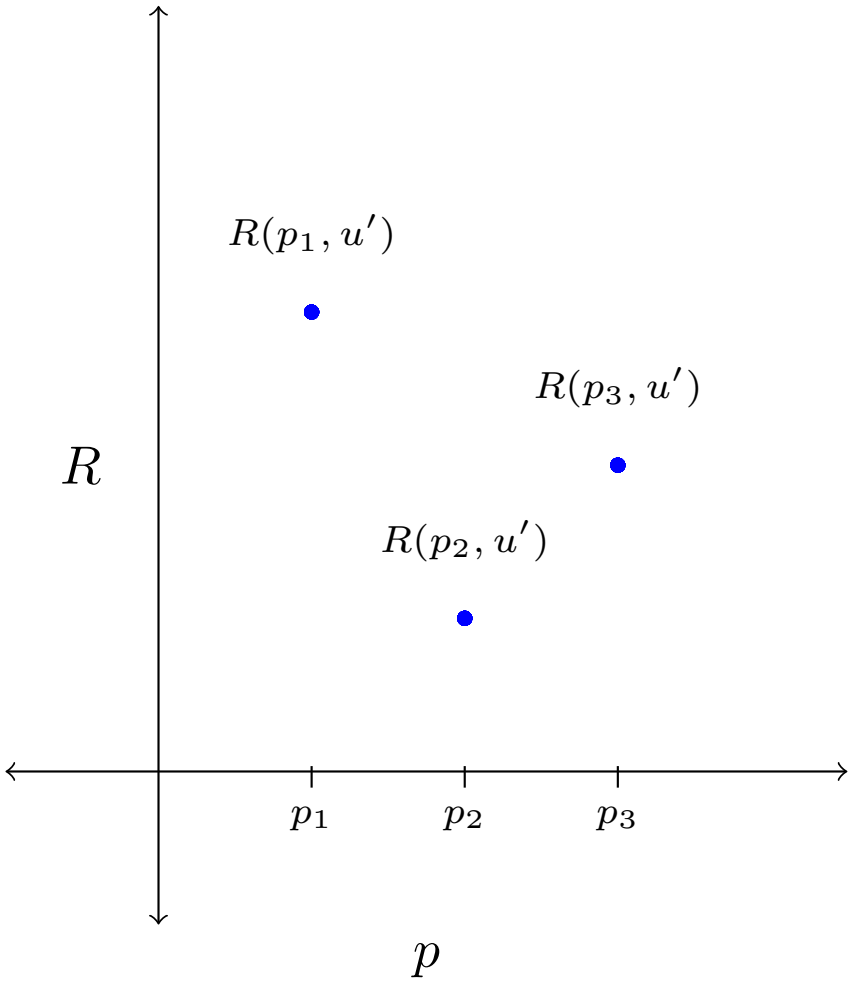
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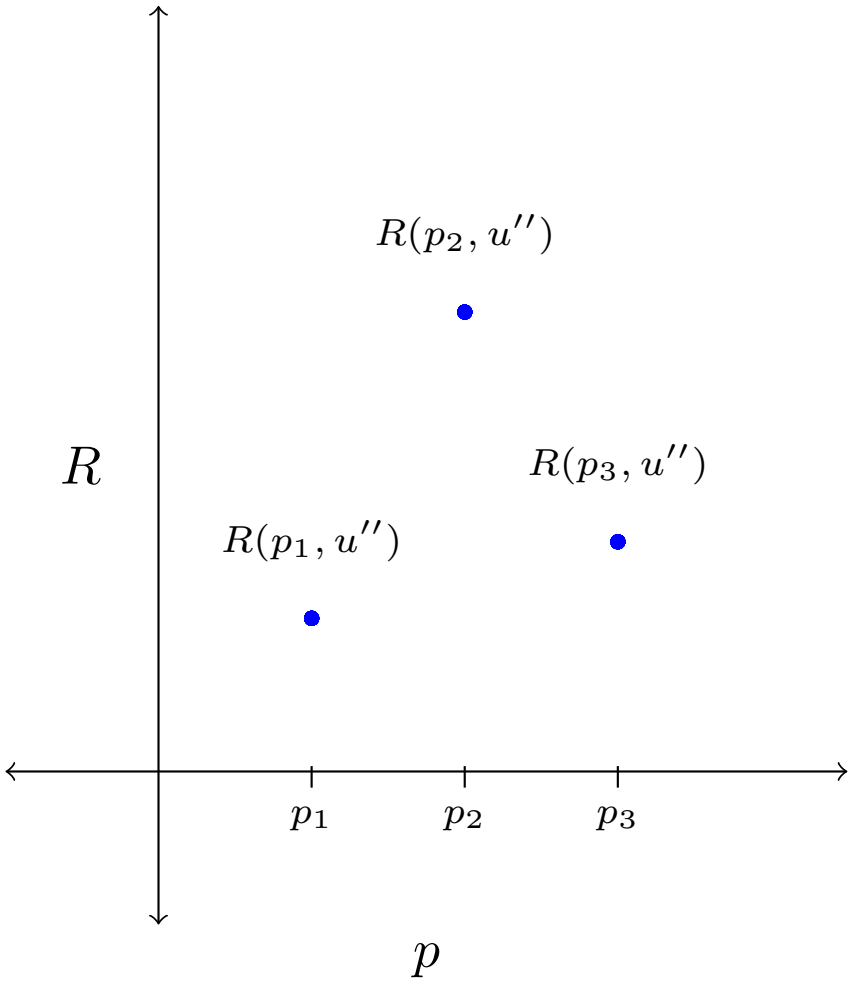
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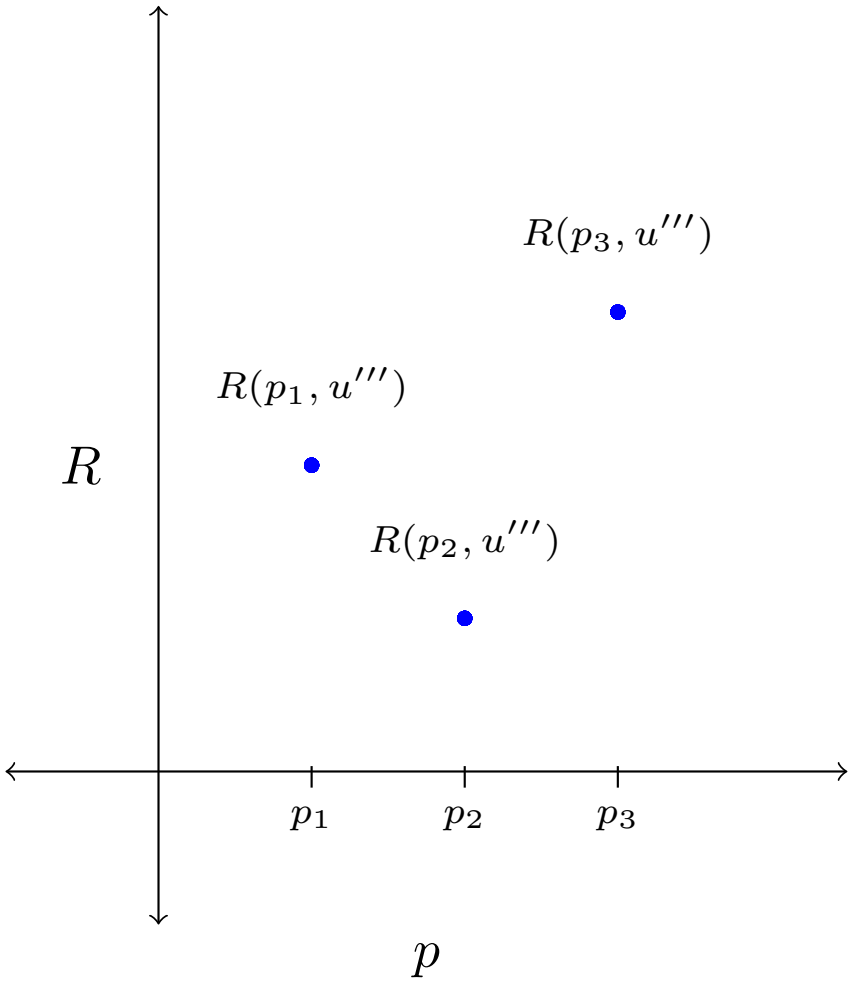
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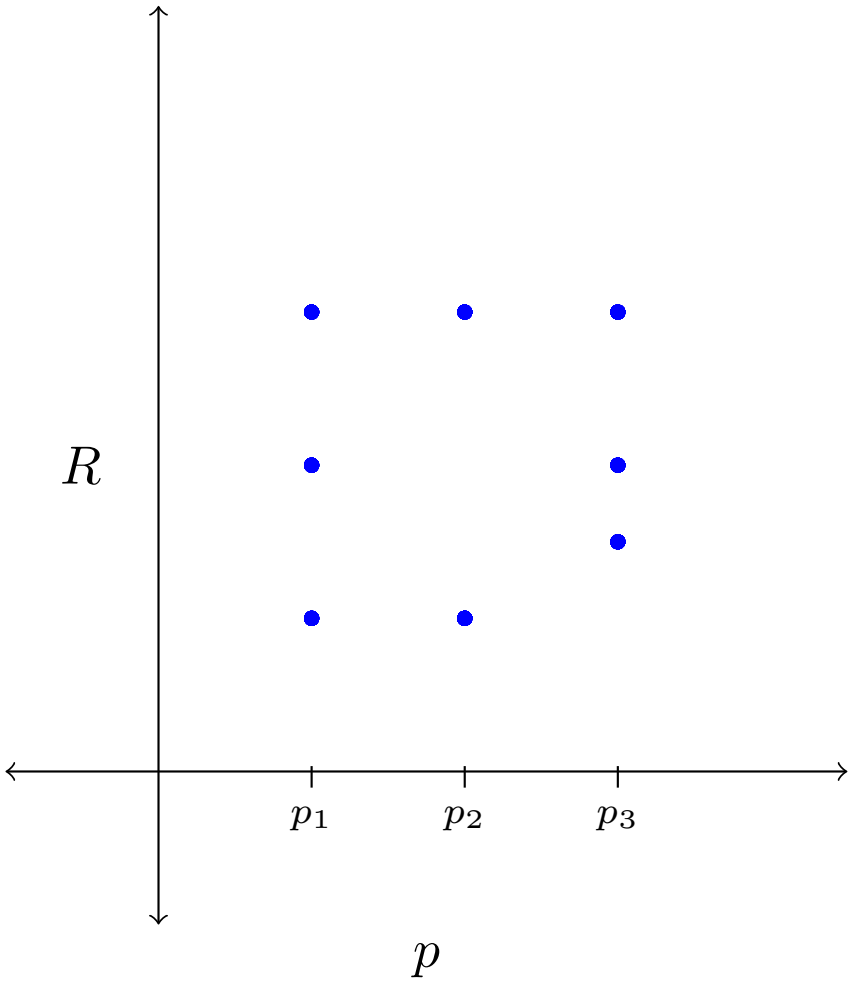
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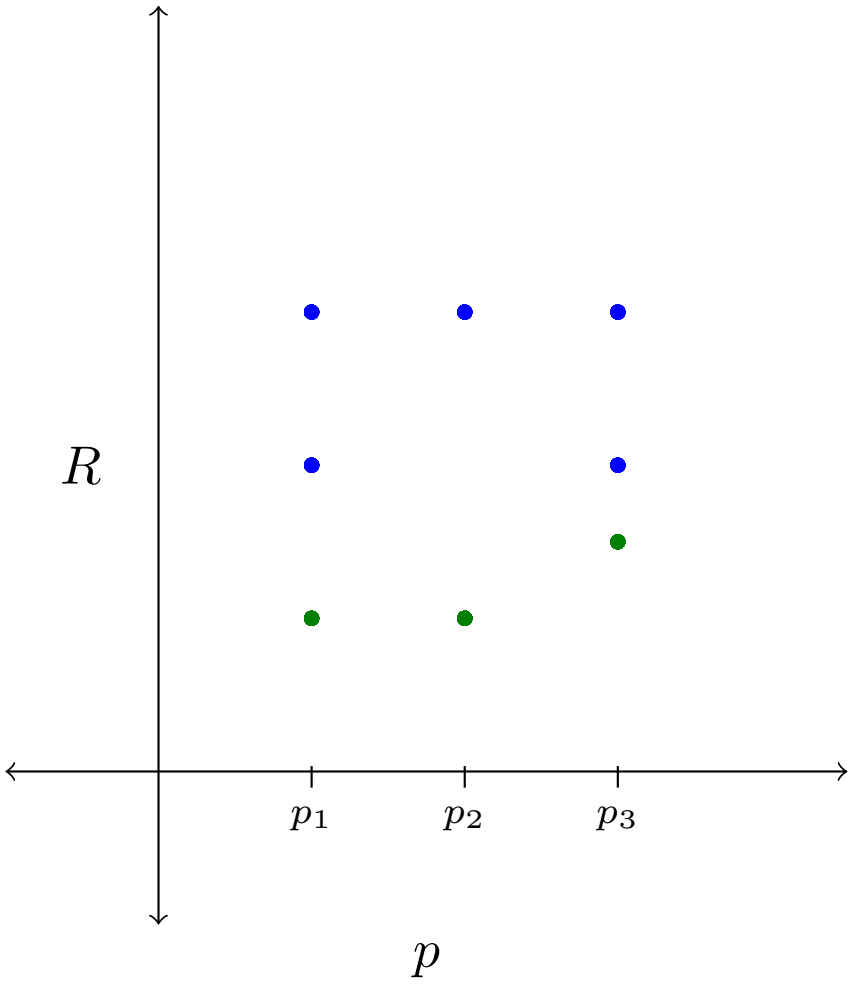
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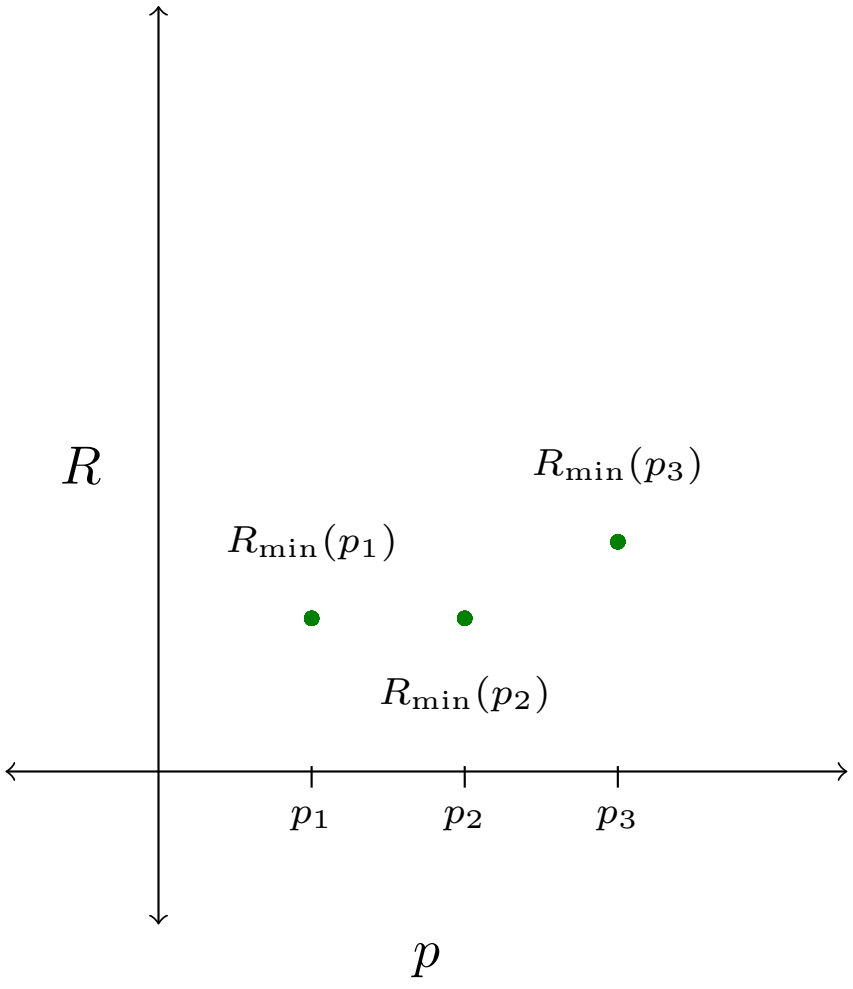
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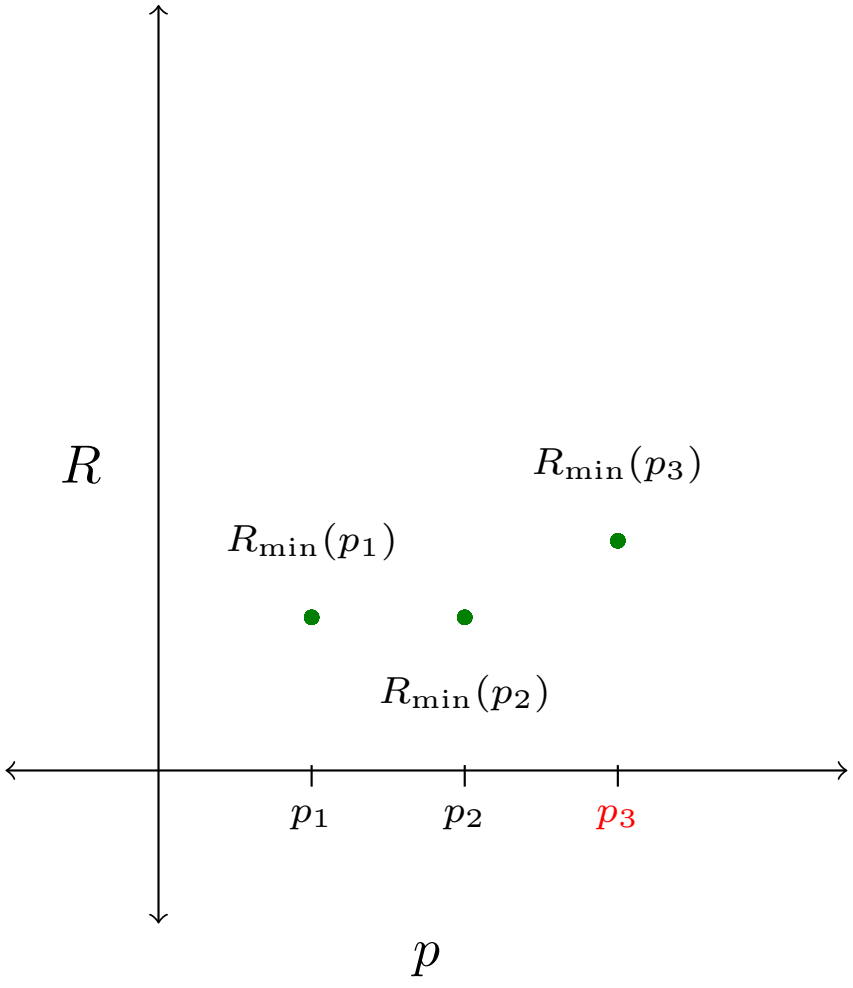
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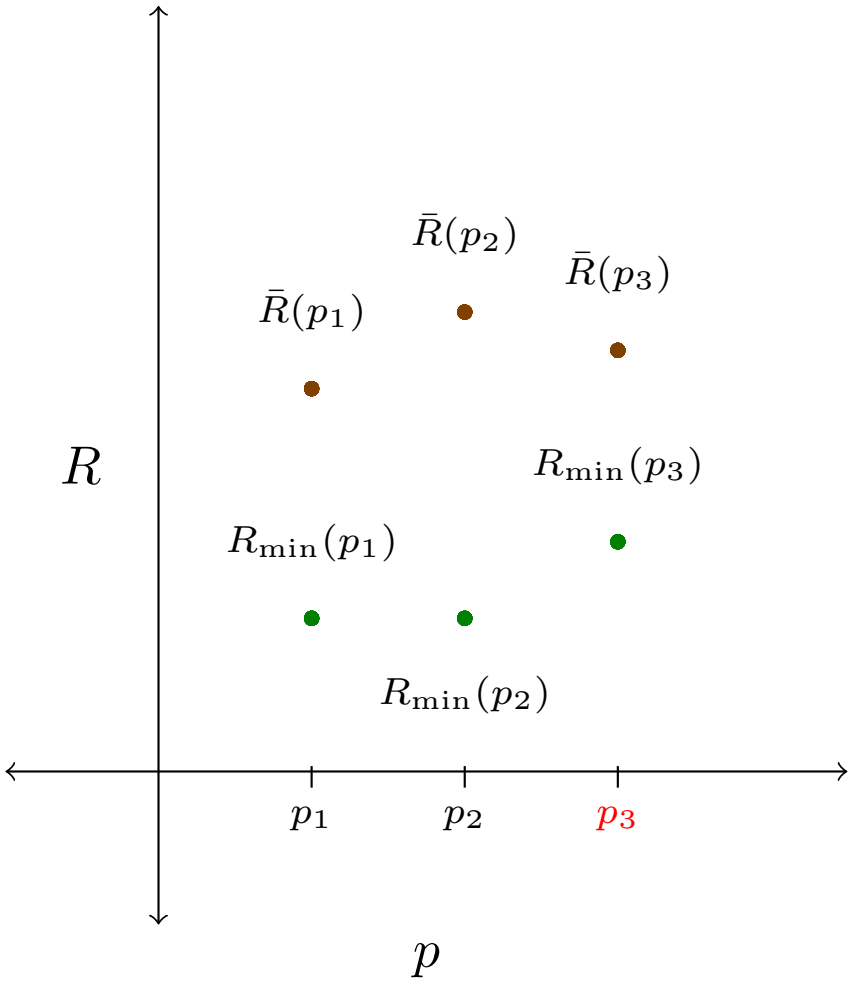
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Suppose we have an uncertainty set $\mathcal{U} = \{u', u'', u'''\}$, price set $\mathcal{P} = \{p_1, p_2, p_3\}$:



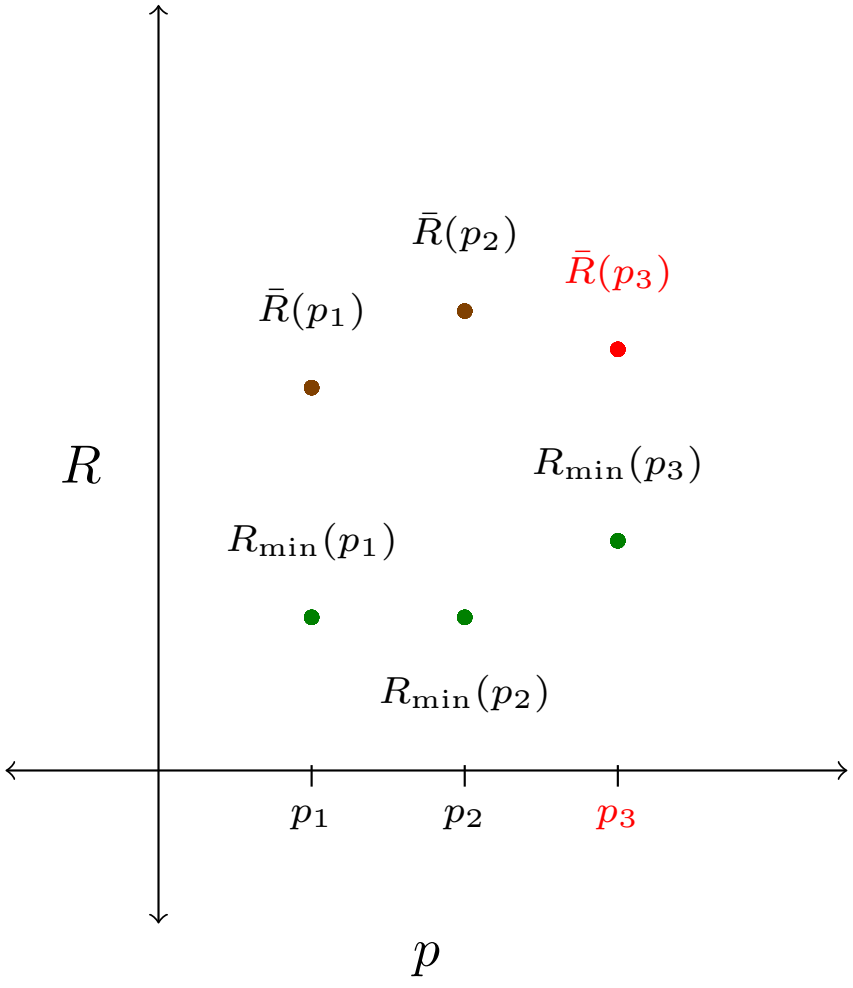
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Restricted pricing problem - example

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Results

Demand model	Type of mean	$\mathrm{RI}_{\mathrm{N}\to\mathrm{RR}}$ (%)	$\mathrm{RI}_{\mathrm{N}\to\mathrm{DR}}$ (%)	$\mathrm{RI}_{\mathrm{N}\to\mathrm{H}}$ (%)
Semi-log	Mean	4.65	1.51	3.09
Semi-log	Trimmed Mean (0.05)	1.76	0.17	1.20
Semi-log	Trimmed Mean (0.10)	1.40	0.21	1.08
Log-log	Mean	11.88	14.31	16.61
Log-log	Trimmed Mean (0.05)	8.52	8.53	11.07
Log-log	Trimmed Mean (0.10)	5.37	3.80	6.57